PEARL: Personalized Streaming Video Understanding Model

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Research Questions

- How can vision-language models handle personalized streaming video understanding with dynamically defined concepts in real-time?
- What benchmark and evaluation framework is needed to assess models' ability to recognize and reason about user-specific concepts across continuous video streams?
- How can a training-free framework enable existing VLMs to perform personalized streaming video understanding without parameter updates?

Core Idea

Current multimodal personalization methods are limited to static images or offline videos, disconnecting them from real-time streaming scenarios needed for AI assistants. The key insight is to decouple concept-centric knowledge from stream-centric observations using a dual-grained memory system that stores both user-defined concepts and historical video clips separately. PEARL works by maintaining these two memory banks and using concept-aware retrieval to find relevant historical evidence when answering queries, enabling real-time personalized responses without retraining.

Key Formula

PSVU Response Generation

$$A = \mathcal{M}(\mathcal{C}_{sub}, \mathcal{V}_{context}, Q)$$

A is the generated response, M is the model, C_sub is the query-relevant concept subset retrieved from concept memory, V_context is the necessary visual context from streaming memory, and Q is the user query. This formula captures how PEARL dynamically constructs context from both concept and streaming memories to generate personalized responses.

Key Figure

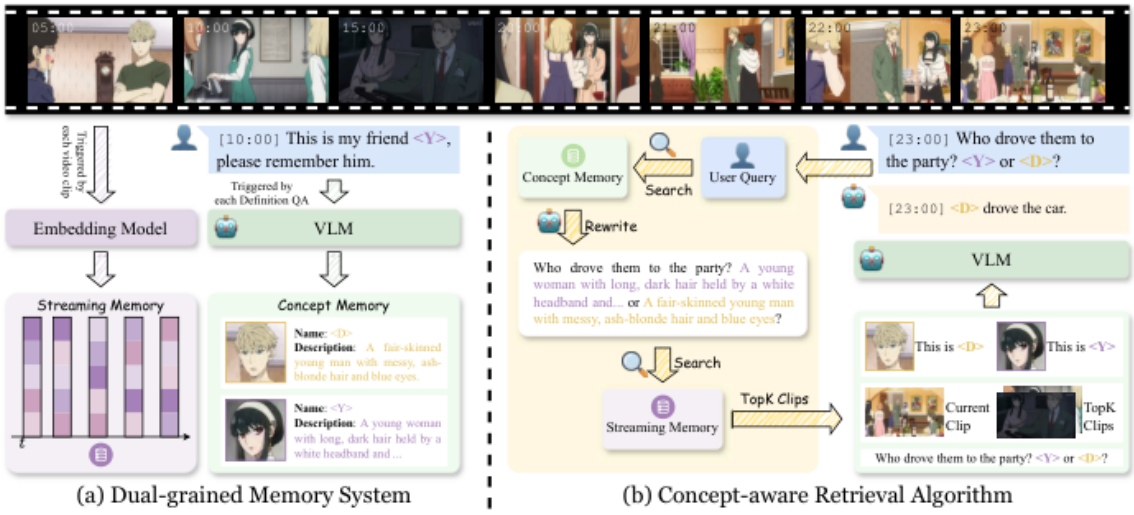

Fig. 3: PEARL framework. (a) Dual-grained Memory System: *Concept Memory* stores user-defined concepts with visual evidence and textual descriptions; *Streaming Memory* archives segmented clips with multimodal embeddings. **(b) Concept-aware Retrieval Algorithm:** Upon a user query, PEARL retrieves relevant concepts and top-K historical clips via concept-rewritten query embeddings, then feeds them together with the current clip into a VLM for real-time personalized response.

an embedding $\mathbf{e}_i = f_{\text{emb}}(\mathcal{X}_i)$, and store the pair $(\mathcal{X}_i, \mathbf{e}_i)$ in Streaming Memory. Each clip embedding $\mathbf{e}_i$ captures rich semantic information about the scene and is used for subsequent retrieval. Detailed settings are provided in the appendix.

Concept Memory When a Concept-Definition query Q_{def} is issued at timestamp t_c, the model creates a new entry with three components: (i) a concept name, (ii) associated visual evidence, and (iii) a textual description. The model first invokes an external tool to extract the visual evidence from the current clip $\mathcal{X}_{t_c}$: for video-level concepts, the evidence is the clip $\mathcal{X}_{t_c}$ itself, whereas for frame-level concepts the model stores the last frame of $\mathcal{X}_{t_c}$. Conditioned on this extracted visual evidence, the model then generates a compact description that summarizes the concept's salient characteristics, using a standardized prompting template provided in the appendix. The resulting entry is finally inserted into the Concept Memory for subsequent retrieval and querying.

4.3 Concept-aware Retrieval Algorithm

When a user issues a Real-Time query or a Past-Time query at timestamp t_q, the model needs three types of information to respond accurately: (i) the query Q, (ii) the query-relevant concept subset $\mathcal{C}_{sub} \subseteq \mathcal{C}$ retrieved from Concept Memory, and (iii) the visual evidence $\mathcal{V}_{context}$ retrieved from Streaming Memory, producing the final answer as $A = \mathcal{M}(\mathcal{C}_{sub}, \mathcal{V}_{context}, Q)$. To obtain $\mathcal{C}_{sub}$, we identify the concept names mentioned in Q and use them as keys to retrieve the corresponding Concept Memory entries. To obtain $\mathcal{V}_{context}$, we use the model $\mathcal{M}$ to rewrite the query into $\tilde{Q}$ by replacing each concept name with its associated description,

Figure 3: Shows PEARL's complete architecture with dual-grained memory system and concept-aware retrieval algorithm,

illustrating how concept memory stores user-defined concepts while streaming memory archives video clips, and how retrieval works for personalized responses.

Limitations

- Requires scene boundary detection and video segmentation which may fail on complex or rapidly changing scenes
- Performance depends on quality of multimodal embedding model for concept-aware retrieval
- Limited to predefined hyperparameters (K=4 clips, N=1 adjacent clips) that may not generalize across all video types
- Concept descriptions generated automatically may not capture all nuanced visual characteristics needed for accurate retrieval